

FLEA NEWS 23

Department of Entomology
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Ames, Iowa 50011 U.S.A.

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Compiled by Robert E. and Joanne H. Lewis

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Please find herewith:

- lists of additional literature for 1976 (list 11), 1977 (list 9), 1978 (list 10), 1979 (list 9), 1980 (list 5), and 1981 (list 2)
- additions and changes to the mailing list

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Thank you for your continued interest in Flea News, your comments and suggestions. In a few instances, Flea News 22 (which was mailed in June 1981) did not reach its destination. Please let us know if you failed to receive your copy.

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It is with deep regret that we record the death of two of our esteemed colleagues, Dr. A. Barrera and Dr. O. I. Scalón. The Universidad Nacional Autónoma de México has sent Dr. Barrera's obituary and bibliography, which are reprinted in this issue.

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JUST PUBLISHED. An illustrated catalogue of the Rothschild collection of fleas (Siphonaptera) in the British Museum (Natural History). Volume VI. Pygiopsyllidae by D. K. Mardon, 1981, viii + 298 pp., 748 figures, £ 50.00. While we have not yet seen this volume, the announcement lists the following contents: Introduction; Scope of volume VI; Taxonomic limitations; Terminology; Synonymies; Listing the material; Type specimens of species and subspecies; List of institutions in which types are deposited; List of collectors' names; Host names; Supplementary glossary; Acknowledgements; Key to the genera of Pygiopsyllidae; Subfamily Lycopsyllinae; Subfamily Pygiopsyllinae; Subfamily Stivaliinae; Index to genera, species and subspecies of Pygiopsyllidae; References; Captions to figures. This and earlier volumes may be obtained from: Publications Sales, British Museum (Natural History), Cromwell Road, London SW7 5BD, England. Payment with order requested.

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PROFESSOR J.-C. Beaucournu, Rennes, has indicated that he is interested in obtaining unpublished records of the flea fauna of the western Mediterranean region. This includes Spain, Portugal, France, Italy, Algeria, Tunisia, and Morocco, as well as the western islands. Your cooperation will be appreciated.

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ADDITIONAL NEW TAXA - In Flea News 22, p. 2, is a partial listing of flea taxa described in 1980 and 1981. We herewith update that list to include additional new taxa that we have learned about since that time. This brings the number described in 1980 to 22, and the number described in 1981 to 4. If we have missed your new species, please send us a reprint. In Flea News 24, we hope to assign numbers to species described in 1980.

1980

Macrostylophora paoshanensis Li & Yan
Ceratophyllus nanshanensis Tsai, Pan & Liu
Cratynius (Angustus) yunnanus Li, Xie & Liao
Mesopsylla hebes clara Smit
Hystrihopsylla rotundisinuata Li & Hsieh
Ctenophthalmus eothenomus Li & Huang
Ctenophthalmus breviprojiciens Li & Huang
Typhlomyopsylla cavaticus Li & Huang
Amphalius spirataenius diqingensis Li, Xie & Yang
Amphalius spirataenius qinghaiensis Li, Xie & Yang
Macrostylophora luchuensis Huang
Paraceras laxisinus Xie, He & Li
Paraceras menetus Xie, He & Li
Genoneopsylla angustidigita Wu, Wu & Tsai
Genoneopsylla claviprocera Hsieh & Wu
Genoneopsylla bisinuata Liu, Chang & Liu
Vermipsylla gilianensis Wu, Tsai & Liu
Callopsylla kazbegiensis Goncharov
Neopsylla fimbriata Li & Hsieh
Neopsylla elesina Li
Neopsylla megaloba Li
Neopsylla schismatosa Li

1981

Nearctopsylla pfitzeri Benton
Amphipsylla kulkarnii Lewis
Vermipsylla ibexa Zhang & Yu
Araeopsylla faini Beaucournu

* * *

THE FOLLOWING PAPERS relating to fleas are listed in the program of the annual meeting of the Entomological Society of America, San Diego, California, 29 November - 3 December 1981.

Hopla, C. E.

Cliff swallows and flea associations in the Nearctic region.

Rust, M. K. & D. A. Reiersen

Influences of cultural practices on indoor flea control.

Rutledge, L. C.

Tests of repellents against Xenopsylla cheopis (Siphonaptera: Pulicidae).

Schwan, T. G.

An association of sex with the incidence of phoretic mite populations on fleas in Kenya.

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Following is a short article which appeared in the Iowa State University Computation Center Newsletter 15(7): 6-7, July, 1981. It describes a data retrieval system currently being developed at Iowa State University with the assistance of Mrs. Nancee Nilsen.

MY FLEAS HAVE FILES!

A SPIRES Application

One of the "newer" systems available to patrons of the Computation Center is the Stanford Public Information REtrieval System (SPIRES). This system is a generalized data base management system developed at Stanford University. SPIRES permits a definition of custom data bases which incorporates sophisticated input and editing of data, element and record retrieval, and file management statistics, all without the formidable investment in developing time-sharing data base management programs.

SPIRES data sets are kept in files which allow efficient retrieval and update. Even a major change to an application data base, such as adding that unforeseen element or a new index, can generally be done with relative ease to the data base owner. Several common SPIRES application examples, catalogs, mailing lists, bibliographic citations, etc., are available at the Computation Center.

A recent application utilizing SPIRES is presently in use by the Department of Entomology. Developing this application required classification of specimens and cross-referencing synonyms of these classifications. It also required bibliographic data which represents an extensive collection of information about the insect order Siphonaptera (you guessed it, fleas).

Accessible by six indices, the Siphonaptera data base is comprised of two major subfiles: classification data and bibliographic data. The bibliographic data can be accessed using indices which identify author, species identification code, subject keywords (including life stages), Siphonapteran genus and/or species, host genus and/or species, or a synonymy scan. The classification data can be accessed by identification code or Siphonapteran genus and/or species.

In order to get automatic synonymy cross-referencing, the classification subfile is self-indexed on a senior synonym code. This feature enables the data preparer to indicate a synonymous relationship for a particular entry. All subsequent cross-referencing will be formed by SPIRES indexing routines.

The classification data is based on a chronological identification code and synonymy relationships carefully recorded by Dr. Robert Lewis. Over 2,300 classifications have been entered which index bibliographic information. Bibliographic data is eventually expected to include over 11,000 citations. These citations are an international collection maintained by a continuous literature review based on computer and manual searches as well as personal correspondence.

Data retrieval is open to the public, but data base security is provided to prevent action that would change or remove any data, except by the file owner.

One input and three output formats were prepared for each of the subfiles. The output formats include a short format, which yields only the most pertinent

information in the fewest lines possible, a standard format which yields information most commonly required by investigators, and the full format which yields all input elements of each file.

The formats utilize sub-goal processing to capture information from both subfiles. The input format enables the data preparer to enter information for new records using prompting techniques. Updates and merges utilize the standard SPIRES transfer methods.

SPIRES protocols are used to find all citations related to a specific name or specific species. Access is provided to both files using information from the classification file to access the bibliographic file. For example, if a species appears in the literature under more than one name, all citations concerning that species can be retrieved and viewed by accessing any one description. Additionally, this request can be very specific and may include only the primary descriptions of the species contained in the bibliographic file.

A sample retrieval session may include the following:

```
-> ..FIND. SYNONYMY
:PLEASE ENTER ID TO BE SCANNED>> 190201
* SYNONYMS FOUND WERE 190201 OR 190325 OR 189002 *
:DO YOU WANT TO SEE JUST THE PRIMARY CITATIONS?(Y/N) N
-RESULT: 5 CITATION(S)
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A resulting citation entry might appear as:

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George Henry Evans Hopkins, 1952
A note on synonymy in Siphonaptera.
T. Washington Academy of Sciences 42(11): 363-365.
CITED-IDS: 189825 Peromyscopsylla spectabilis
           190201 Trichopsylla ursi
           190325 Vermipsylla strandi
           ...
```

Further information about the data contained in this SPIRES application can be obtained by contacting Dr. Robert Lewis, Department of Entomology, 418 Science II (1-515-294-1815).

SPIRES definitions, formats, and protocols related to the Siphonaptera application can be obtained by contacting Nancee Nilsen, 4 Computer Science (1-515-294-4171).

-- Nancee Nilsen

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MAILING LIST

ADDITIONS:

Entomology Library, Yale Peabody Museum, P.O. Box 6666, New Haven, Connecticut
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Ms Ann Lum, Entomology Library, British Museum (Natural History), Cromwell Road,
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Dr. A. J. Main, Jr., Dept. of Epidemiology & Public Health, Yale University,
P.O. Box 3333, 60 College Street, New Haven, Connecticut 06510 USA

Mr. D. P. Sleeman, Dept. of Agricultural Biology, University College, Dublin,
Belfield, Dublin 4, IRELAND

Dr. J. O. Whitaker, Jr., Dept. of Life Sciences, Indiana State University,
Terre Haute, Indiana 47809 USA

CHANGES:

Dr. H. J. Egoscue, P.O. Box 787, Grantsville, Utah 84029 USA

Dr. H. E. Stark, RFD 286-B, Trenton, Utah 84338 USA

* * *

An asterisk (*) preceding a citation indicates we have seen neither the original publication nor verified it in another source. Anything appearing in brackets ([]) after a citation indicates an annotation, e.g., the name of a taxon described.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1976 (list 11)

*Ranade, D. R. & D. V. Bhalchan - Note on natural infection in the rat flea, *Xenopsylla cheopis* (Roths.) with *Trichinella spiralis* (Owen). J. Commun. Dis. 8(1): 77.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1977 (list 9)

*Isaacs, M. & J. R. Jamison - Epidemiological and clinical aspects of plague in southern Africa. In: J.H.S. Gear (ed.), Proc. intl. Symp. on Medicine in a tropical Environment, Pretoria, S. Africa, 1976, pp. 479-486. Balkema, Cape Town.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1978 (list 10)

*Connor, J. D., R. A. Williams, M. A. Thompson, M. Ginsberg & S. Daley - Plague in San Diego. Western J. Med. 129(5): 394-406. [A specialty conference.]

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1979 (list 9)

*Gilles, H. M. - The spread and control of arthropod-borne diseases. In: W. Hobson (ed.), The Theory and Practice of Public Health, 5th ed., pp. 266-276. Oxford Univ. Press.

Haarløv, N. - Humane ektoparasitter. In: J. Søndergaard (ed.), Dermatologi og Venerologi, pp. 363-383. Christian Ejlers' Forlag, København.

Kieffer, M., S. Kristensen & T. E. Hallas - Prurigo and pets: the benefit from vets. Brit. med. J. No. 6177, vol. 1: 1539-1540.

*Kozlov, M. P. - Chuma (Plague). Published by Meditsina, Moskva, 191 pp., 8 figs., 6 tables.

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1980 (list 5)

*Alexander, J. T. - Vien, Ivan and the 1st comprehensive plague tractate in Russian. Med. Hist. 24(4): 419-431.

Allison, F. R. - Studies of batflies and bat-fleas in Queensland. Australian Bat Res. News 16: 9-10.

(1980)

- Anonymous - Environmental management for vector control. Third Report of WHO Committee on Vector Biology and Control. WHO Tech. Rep. Series 649, pp. 1-75.
- *Anonymous - Myxomatosis [Field trials, virus strains, genetic resistance, rabbit fleas]. CSIRO Div. Wildlife Res. Report 1978-80: 44-46.
- Anonymous - Resistance of vectors of disease to pesticides. Fifth Report of WHO Expert Committee on Vector Biology and Control. WHO Tech. Rep. Series 655, pp. 1-82.
- *Abul-hab, J. - A list of Arthropoda of medical and veterinary importance recorded from Iraq. Bull. biol. Res. Centre, Baghdad 12/1: 9-39.
- *Ata-ur-Rahim, M. - Medical literature on Pakistan from 1840-1947: a bibliography. J. Pakistan med. Assoc. 30(8-10): 189-193; 215-217; 234-236.
- Chin, T.-h. & K.-c. Li - On *Thaumapsylla breviceps orientalis* Smit, 1954 from China. Acta Acad. Med. Guiyang 5(2): 135-136.
- *Comiti, V.-P. Les maladies d'autrefois. Recherche, Paris 115: 1044-1048.
- *Croce, J. - Hipersensibilidade as picadas de pulga. Medico, Porto 97(1525): 572-576.
- Dinger, S., R. Cantoray & E. Taşan - Ectoparasites and endoparasites of street cats of Elazig. Firat Univ. Vet. Fak. Dergisi 5(1): 7-15.
- *Dingelde, W. - Control of ectoparasites in farm animals with a new type of insecticide. Tierzuchter 32(2): 47-48.
- Egoscue, H. J. - Woodrat nest flea *Anomiopsyllus amphibolus* in southeastern Oregon. Great Basin Naturalist 40(4): 361.
- Ell, S. R. - Interhuman transmission of medieval plague. Bull. Hist. Med. 54(4): 497-510.
- *Emelianov, P. F. et al. - On the problem of the role of the Persian gerbil in natural foci of plague. II. Biol. Zh. Armenia 33(12): 1255-1261.
- *Gracio, A. J. S. - The plague in Angola, entomological survey on two outbreaks of bubonic plague in Bocoio (1973/74) and Cuito-Cuanavale (1975). Abstr. 306-307, 10th Intl. Congr. Trop. Med. & Malaria, Manila, Philippines, Nov. 9-19, 1980.
- Haas, G. E., L. Johnson & N. Wilson - Siphonaptera from mammals in Alaska. Supplement II. Southeastern Alaska. J. entomol. Soc. Brit. Columbia 77: 43-46.
- Haitlinger, R. - Arthropoda (Siphonaptera, Anoplura, Coleoptera, Acarina) collected from small mammals in Rumania. Wiad. Parazytol. 26(6): 679-710.
- Huang, G.-p. - A new species of genus *Macrostylophora* from Yunnan (Siphonaptera: Ceratophyllidae). Acta Acad. Med. Guiyang 5(2): 132-134. [Description of *M. luchuensis*]
- Li, K.-c. - On the species-groups, zoogeographical divisions and host-relationships of the Chinese *Macrostylophora* Ewing, 1929 (Siphonaptera: Ceratophyllidae). Acta Acad. Med. Guiyang 5(2): 108-113, 1 table
- Li, K.-c. - Fleas of *Hylomys suillus* of China. Acta Acad. Med. Guiyang 5(2): 128-129.

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- Li, K.-c. - Fleas of *Typhlomys cinereus* of China. Acta Acad. Med. Guiyang 5(2): 130-131.
- Li, K.-c. - Checklist of fleas of Taiwan province. Acta Acad. Med. Guiyang 5(2): 194.
- Li, K.-c. - Siphonaptera collected in Gongshan, Yunnan, II. Entomotaxonomia 2(3): 177-184, figs. 1-7c. [Description of *Neopsylla fimbriata* Li & Hsieh; *Neopsylla elesina* Li; *Neopsylla megaloba* Li; *Neopsylla schismatosa* Li]
- Li, K.-c. & G.-p. Huang - A survey on fleas from Kuankuoshui, a natural reserve in Guizhou Province. Acta Acad. Med. Guiyang 5(2): 114-122, figs. 1-5. [Description of *Ctenophthalmus eothemus*; *Ctenophthalmus breviprojiciens*; *Typhlomopsylla cavaticus*]
- Li, K.-c., B.-q. Xie & H.-r. Liao - A new species and subgenus of the genus *Cratynius* Jordan, 1933 (Siphonaptera: Ceratophyllidae). Entomotaxonomia 2(1): 41-45, figs. 1-7. [Description of new subgenus (*Angustus*); new species *yunnanus*]
- Li, K.-c., B.-q. Xie & X.-d. Yang - A new subspecies of *Amphalius spirataenius* Liu, Wu & Wu, 1966, with a discussion on its subspecies in China. (Siphonaptera: Ceratophyllidae). Acta Acad. Med. Guiyang 5(2): 123-127, figs. 1-3. [Description of new subspecies *diqingensis*; *qinghaiensis*]
- *Lloyd, H. G. - The red fox, with reference to population dynamics, methods of control, parasites and diseases. B. T. Batsford, London. pp. 320.
- Rubtsov, I. A. - Symbiogenesis of entomogenous helminths with dipterans and fleas. Zh. obshch. Biol. 41(6): 845-859.
- *Service, M. W. - A guide to medical entomology. MacMillan. pp. 208, 200 figs.
- *Shawkey, K. E. - The epidemiological pattern of sylvatic plague in the steppes of Libya. Abstr. 307-308; 10th International Congress of Tropical Medicine & Malaria, Manila, Philippines, Nov. 9-19, 1980.
- Smith, S. A. & M. E. Clay - Morphology of the Bat Flea, *Myodopsylla insignis* (Siphonaptera: Ischnopsyllidae) with special reference to the pygidium. Ohio J. Sci. 80: 12 [Abstr. of paper presented at 89th annual meeting, Ohio Acad. Sci., April 18-20, 1980]
- Solomon, L. - Aspects about the parasitocoenosis of some small mammals from three phytocoenosis types. Anal. stiint. univ. "Al. I. Cuza" din Iasi (Ser. noua), sect. II (Biol.) 26: 83-92.
- *Splister, H. & G. Ilchmann - Strategy of ectoparasite control in extensive animal farming. Beitr. z. Trop. Landwirtsch. & Vet.-med. 18(3): 235-243.
- *Van der Zee, D., N. P. Ngwaxaxa & M. Mjoli - A survey of Siphonaptera in the Tyume Valley. Fort Hare Papers 7(2): 132-137, 1 fig.
- Xie, B.-q., J.-h. He, R.-h. Chen & K.-c. Li - On two new species of *Paraceras* Wagner, from southwest China (Siphonaptera, Ceratophyllidae). Acta Acad. Med. Guiyang 5(2): 137-144, figs. 1-20. [Description of *P. laxisinus* Xie, He & Li, and *P. menetus* Xie, Chen & Li]

LITERATURE ON SIPHONAPTERA PUBLISHED IN 1981 (list 2)

- Anonymous - Siphonaptera. Zool. Rec. [1976] 113(13a): 232-235.
- Beaucournu, J. C. - *Araeopsylla faini* n. sp. Puce nouvelle du Rwanda (Siphonaptera, *Ischnopsyllidae*). Rev. Zool. afr. 95(1): 195-199, 4 figs.
- Bernath, R. F. & T. H. Kunz - Structure and dynamics of arthropod communities in bat guano deposits in buildings. Canad. J. Zool. 59(2): 260-270.
- Bogdanov, I. I. - Fleas (Siphonaptera) from the Taimyr peninsula. Parazitologiya 15(3): 293-295.
- Cerny, V. - Book review of Kozlov, M.P., 1979, Chuma (Plague), Meditsina, Moskva, 191 pp., 8 figs., 6 tables. Folia parasitol. 28(1): 59. [in English]
- *Kilonzo, B. S., N. R. Patel & R. S. Mtoi - Studies on the seasonal fluctuations of rodents and their fleas in North-Eastern Tanzania. Tanzanian Vet. Bull. 3(1): 3-19.
- Morozov, Yu. A. & U. A. Sagimbekov - The capacity of *Xenopsylla gerbilli minax* and *X. conformis* fleas to feed on man. Med. Parazit., Moskva 50(3): 62-64.
- Nechaeva, L. K., A. N. Kozlov & I. F. Zhovtyi - Study of ectoparasites of the brown rat (*Rattus norvegicus norvegicus* Berk., 1769) in northern Kazakhstan. Med. Parazit., Moskva 50(3): 69-74.
- Nikitina, N. A. & G. A. Nikolaeva - A study of the ability of rodents to clean themselves of specific and non-specific flea species. Zool.Zh. 60(1): 165-167.
- Nikulina, N. A. & T. P. Romasheva - On fleas (Siphonaptera) of *Sorex daphaenodon* Thomas from Charskaya Hollow. Parazitologiya 15(4): 364-367.
- Rubzow, U. A. - New genera and a new species of mermithids (Nematoda, *Mermithidae*) infecting fleas. Parazitologiya 15(4): 338-341.
- Samurov, M. A. - On the effect of allantonematodes on the reproduction and abundance of the flea *Ceratophyllus laeviceps* (Siphonaptera). Parazitologiya 15(4): 360-363.
- Scalon, O. I. - On fleas from Eastern Mongolia with a description of the male and female of *Echidnophaga tiscadaea* Smit, 1967 (Siphonaptera). Parazitologiya 15(3): 280-287, 3 figs., 1 table.
- Shevchenko, V. L. - On migration of *Frontopsylla frontalis alatau* Fed., 1946, fleas of the isabelline wheatear (*Oenanthe isabellina* Temm., 1829). Med. Parazit., Moskva 50(3): 64-69.
- Sleeman, D. P. - Fleas from Irish mammals. Ir. Nat. J. 20(6): 244-247.
- Smit, F. G. A. M. - The song of a flea.--A stridulating mechanism in Siphonaptera? Ent. Scand. Suppl. 15: 171-172, figs. 1-4.
- Soldatkin, I. S. & Yu. V. Rudenchik - On some methodological problems of evaluation of rodent eradication as a preventive measure in plague control. Med. Parazit., Moskva 50(3): 44-48.
- Townsend, D. - New "Pill" stops flea reproduction. Copyright 1981 Los Angeles Times; reprinted Des Moines [Iowa] Sunday Register, Nov. 8, 1981.

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It will be understood that - although their titles are given in English in these lists - papers in Russian or Chinese periodicals, or books published in the USSR or China, are in Russian or Chinese (but often with summaries in English).

Colleagues at the Universidad Nacional Autónoma de México have kindly furnished us with an obituary and bibliography for the late Dr. Barrera. Below is an English translation adapted from the Spanish by Mr. Isaac M. Ortega and Dr. William Franklin, Department of Animal Ecology, Iowa State University. On page 10 we have reprinted the original obituary in Spanish. The bibliography has been restructured to fit the style used by Mr. Smit in the bibliographic sections of earlier issues of Flea News. We gratefully acknowledge the assistance of Dr. Robert Traub and Dr. Nixon Wilson in making the list as complete as possible.

OBITUARY

Dr. ALFREDO BARRERA VÁZQUEZ MARÍN 11 May 1926 - 7 December 1980

Dr. Alfredo Barrera V. Marín, Professor on the Faculty of Sciences of the National Autonomous University of Mexico, died at the age of 54, on 7 December 1980, in the city of Merida, Yucatan, Mexico.

A specialist in ectoparasitic insects, principally the Siphonaptera, he taught and directed research for over thirty years in several institutions in Mexico. During this period he published more than eighty articles on various subjects.

He studied at the National School of Biological Sciences of the National Polytechnic Institute (IPN). Later he traveled to several foreign universities including: the University of Texas, Austin, TX in 1963; the National Pedagogic Institute, Paris, France, in 1963; and the University of Maryland, Baltimore, MD during 1964 and 1965.

During his academic life he received several distinctions. In 1958 and 1959 he was Vice President of the Natural History Society of Mexico. In 1961 he was Vice President of the Entomological Society of Mexico. During 1964 and 1965 he was President of the Natural History Society of Mexico. From 1969 to 1970 he was Vice President of the Scientific Research Academy and in 1970 to 1971, President of the same. From 1973 he was a Correspondent Member of the Natural Sciences Institute of Ecuador. From 1978 to 1980 he was a member of the Review Commission of the Science Faculty. He was also a member of the Postgraduate Study Council of the same institution.

For ten years (1965-1974) he was the Director of the Natural History Museum of Mexico City.

He was a professor and researcher in several institutions in Mexico, among them: Secretary of Health and Assistance, Secretary of Agriculture and Livestock, the National Polytechnic Institute and the National Autonomous University of Mexico.

The merit he enjoyed as a scholar and a teacher was great, his teachings being of inestimable value to his students.

Obituario.

Dr. Alfredo Barrera Vázquez Marín.

El Dr. Alfredo Barrera V. Marín, profesor de tiempo completo de la Facultad de Ciencias de la Universidad Nacional Autónoma de México, murió a la edad de 54 años el 7 de diciembre de 1980 en la ciudad de Mérida, Yucatán, México.

Especialista en insectos ectoparásitos, principalmente sifonapteros, enseñó e investigó durante más de treinta años en diversas instituciones de México, tiempo durante el cual publicó más de 80 artículos sobre el tema.

Hizo sus estudios en la Escuela Nacional de Ciencias Biológicas del IPN, realizando posteriormente varios viajes a diversas universidades del extranjero, entre ellas: la Universidad de Texas, Austin Texas, USA en 1963; el Instituto Pedagógico Nacional, París Francia en 1963; y la Universidad de Maryland, Baltimore USA, durante 1964 y 1965.

Durante su vida académica recibió varias distinciones. En 1958 y 1959 fue Vicepresidente de la Sociedad Mexicana de Historia Natural; en 1961, vicepresidente de la Sociedad Mexicana de Entomología; durante 1964 y 1965 Presidente de la Sociedad Mexicana de Historia Natural; de 1969 a 1970, Vicepresidente de la Academia de la Investigación Científica y de 1970 a 1971, Presidente de la misma; desde 1973 fue Miembro Correspondiente del Instituto Ecuatoriano de Ciencias Naturales; y de 1978 a 1980, Miembro de la Comisión Dictaminadora de la Facultad de Ciencias, así como Miembro del Consejo de Estudios de Postgrado de esta misma.

Durante casi diez años (1965-1974) fungió como Director del Museo de Historia Natural de la Ciudad de México.

Fue maestro e investigador de diversas instituciones del país, entre ellas: la Secretaría de Salubridad y Asistencia, la Secretaría de Agricultura y Ganadería, el Instituto Politécnico Nacional (IPN) y la Universidad Nacional Autónoma de México.

El mérito que gozó como estudioso y maestro fue grande, siendo sus enseñanzas de un valor inapreciable para sus alumnos.